WEB-BASED DOCUMENT TRACKING AND MANAGEMENT PLATFORM

for Secure and Efficient Document Workflow Automation

A THESIS FINAL PRESENTATION

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STATEMENT OF THE PROBLEM

Why the system was developed

MANUAL ROUTING

Physical delivery and handwritten tracking make status monitoring slow and dependent on personnel availability.

LIMITED VISIBILITY

Users cannot consistently see the current holder, pending action, route history, or document status in real time.

RECORD RISKS

Paper files and disconnected logs increase the risk of delays, misplacement, duplicate entries, and incomplete accountability.

Core research question: How can BLGF Region II improve the security and efficiency of its document workflow through a web-based platform?

OBJECTIVES OF THE STUDY

GENERAL OBJECTIVE

Develop and evaluate a Web-Based Document Tracking and Management Platform for secure and efficient document workflow automation at BLGF Region II.

- Document the problems encountered in the existing manual process.
- Design and implement registration, routing, tracking, search, notifications, reporting, and audit functions.
- Evaluate the developed system using the eight ISO/IEC 25010 product-quality characteristics.
- Provide conclusions and recommendations based on actual evaluation results.

SCOPE AND DELIMITATION

IN SCOPE

Incoming and outgoing documents
Real-time routing and status tracking
Role and permission controls
Attachments and recipient conversations
Search, notifications, QR, slips and reports
Audit and envelope dispatch logs

DELIMITATIONS

Designed for BLGF Region II workflows
Authorized users and supported web environment
Evaluation based on 23 respondents
27 indicators under ISO/IEC 25010
Deployment and policies remain subject to agency approval

RESEARCH METHODOLOGY

DESIGN

Descriptive research identified workflow problems; developmental research guided the design, construction, testing, and evaluation of the WDTMP.

PARTICIPANTS

23 evaluators
13 BLGF employees
10 IT experts

INSTRUMENT

A 27-indicator questionnaire covering all eight ISO/IEC 25010 product-quality characteristics, rated through a five-point Likert scale.

Data analysis: weighted means per indicator and category; 4.21-5.00 = Very Great Extent.

DEVELOPMENT APPROACH

Iterative Agile / controlled overlapping phases



Feedback was used to refine routing, permissions, attachments, notifications, reports, and usability before succeeding increments.

SYSTEM ARCHITECTURE

Presentation, application, and data tiers

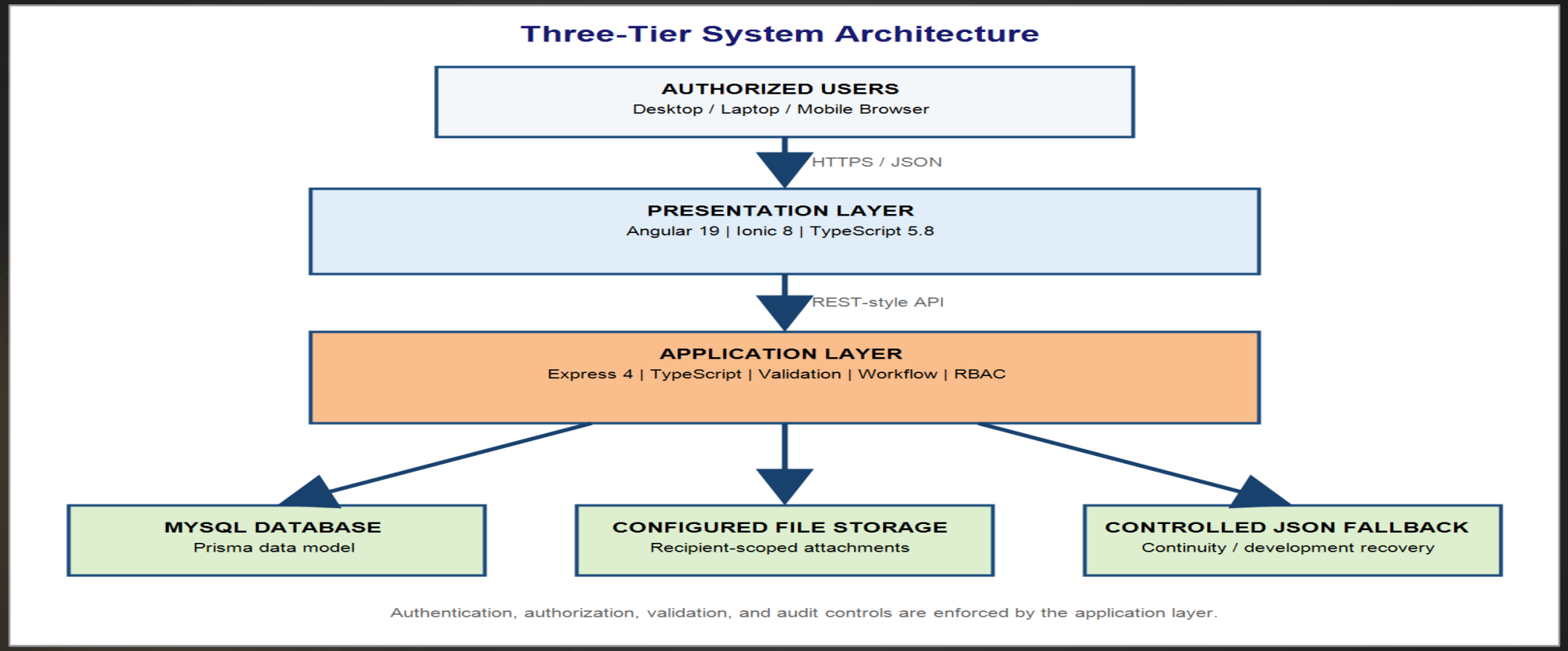


Figure 2. Multi-tier architecture of the WDTMP

CONTEXT-LEVEL DATA FLOW

System boundary and external information exchanges

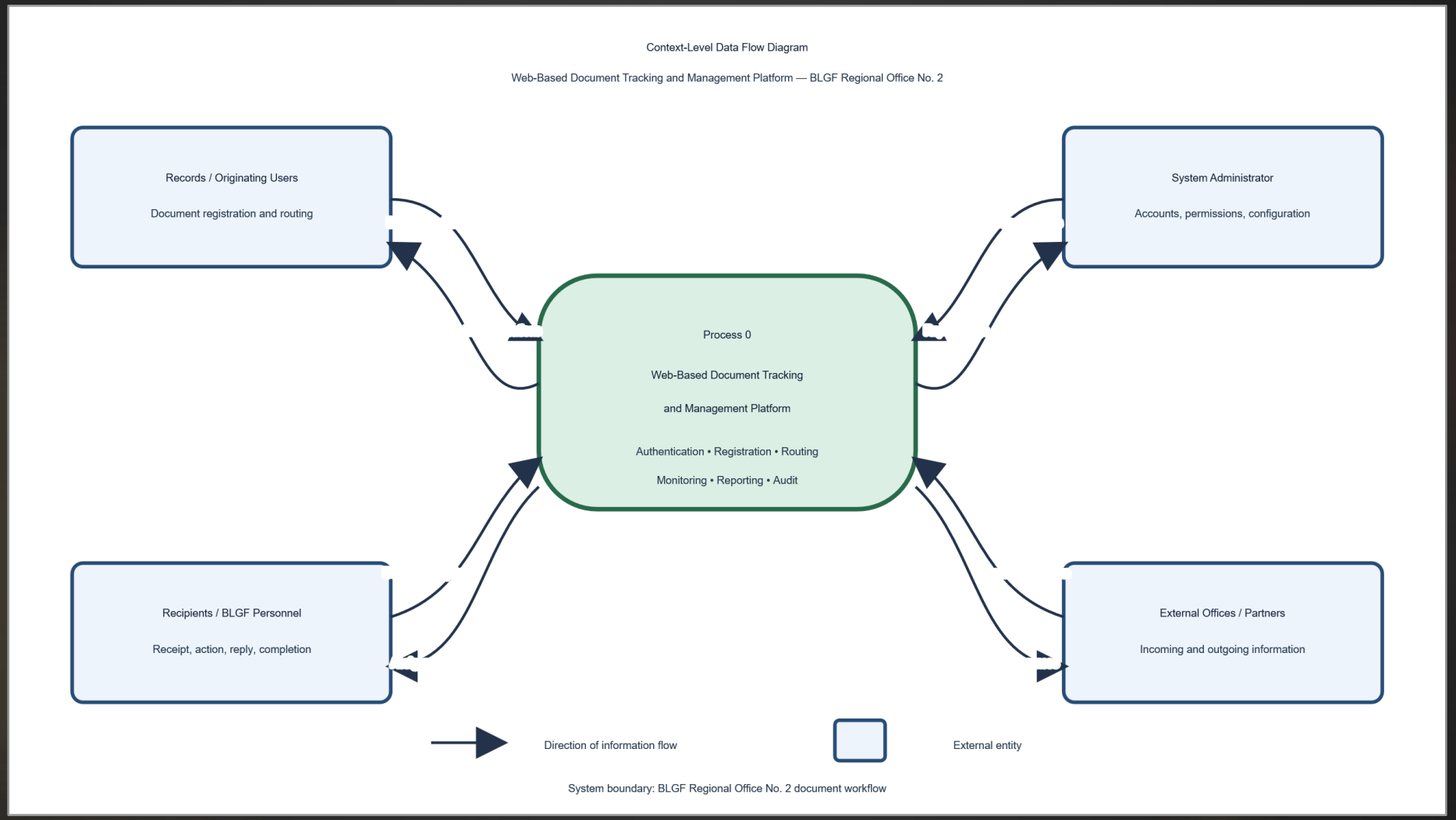
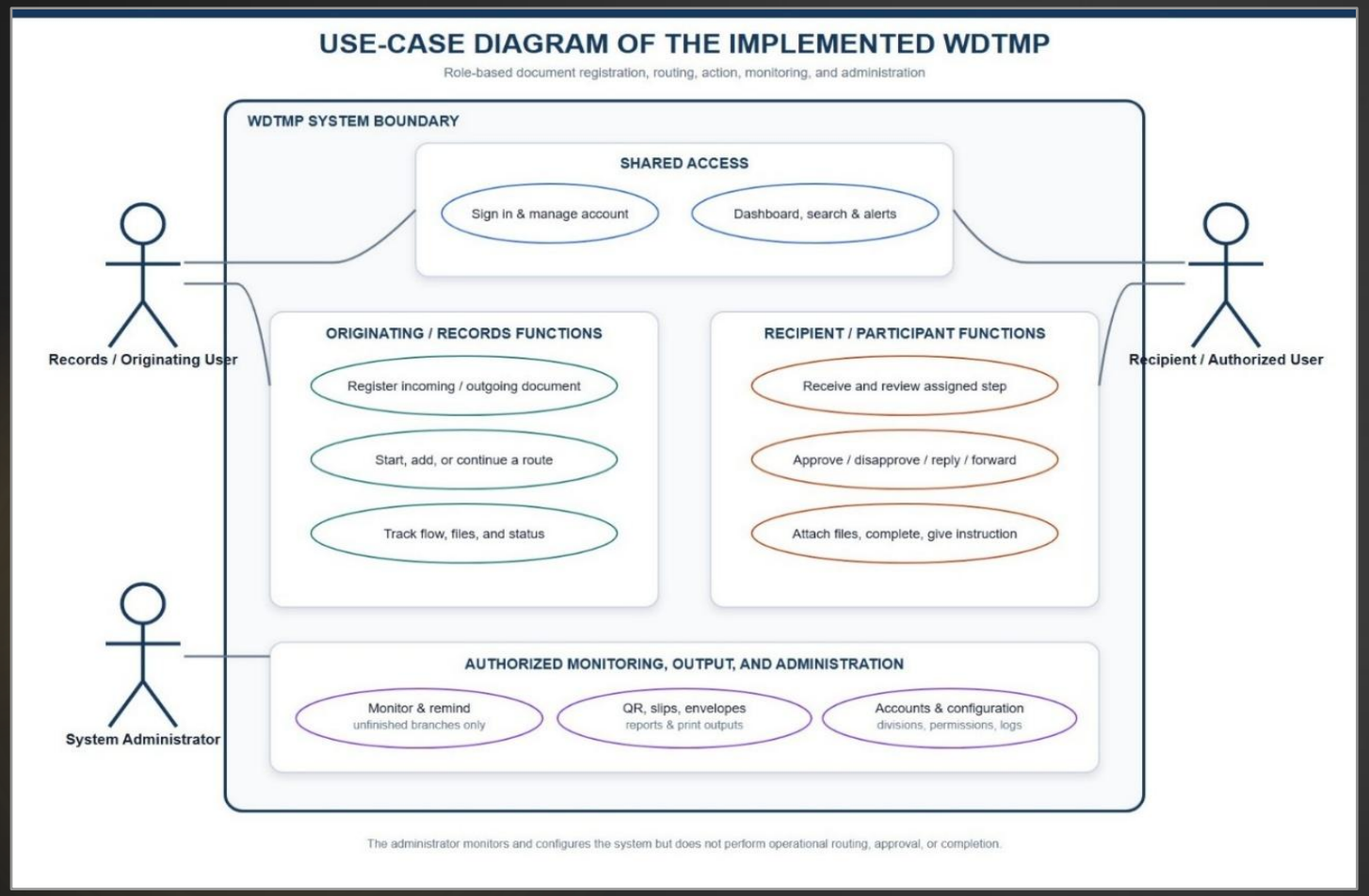


Figure 4. Context-Level Data Flow Diagram

ACTORS AND CORE WORKFLOW

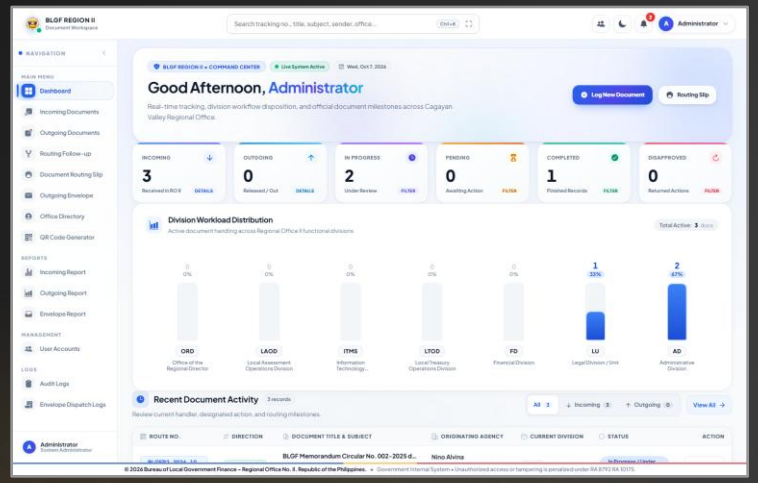


Implemented use-case diagram

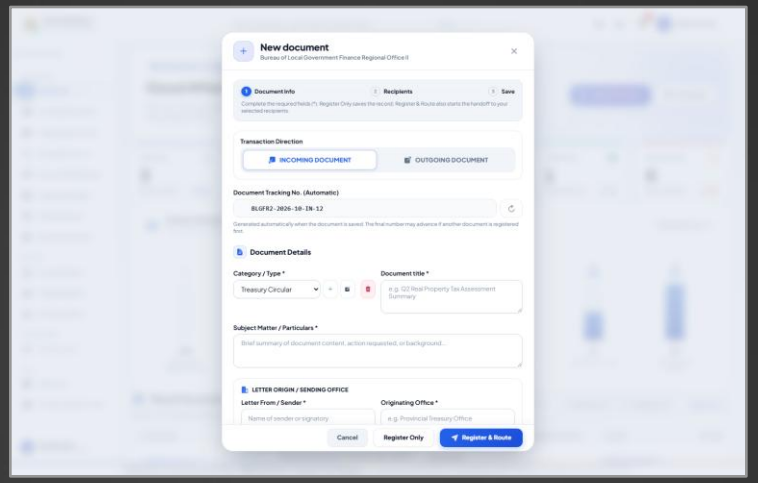
- System Administrator: accounts, permissions, settings, reports and logs
- Records/Originating User: registers and initiates document routes
- Recipient/Authorized User: receives, acts, replies, attaches and completes
- Every significant action remains traceable through route history and audit logs

SYSTEM DEMONSTRATION I

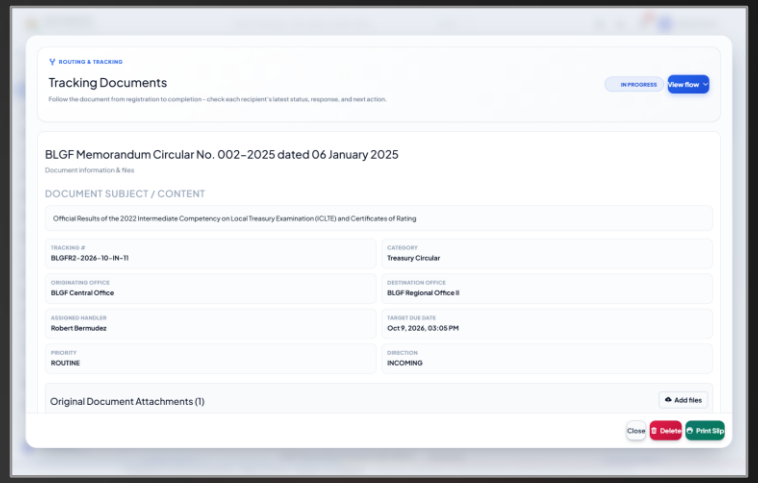
Registration, dashboard, and document routing



Dashboard



Log New Document

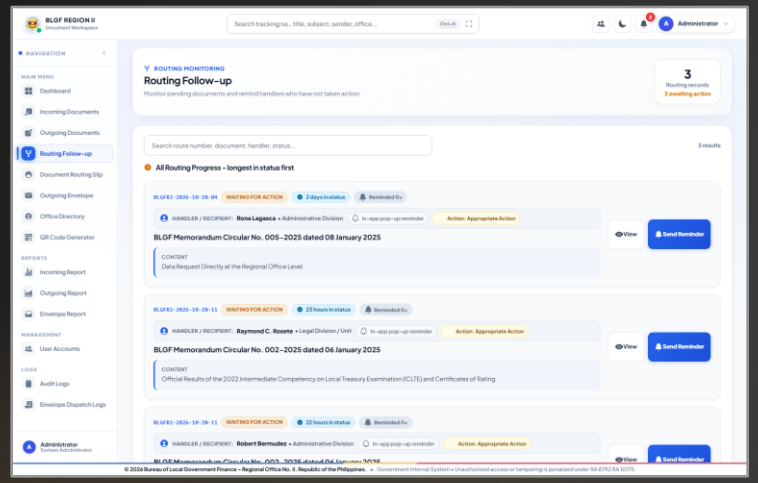


Details and Routing

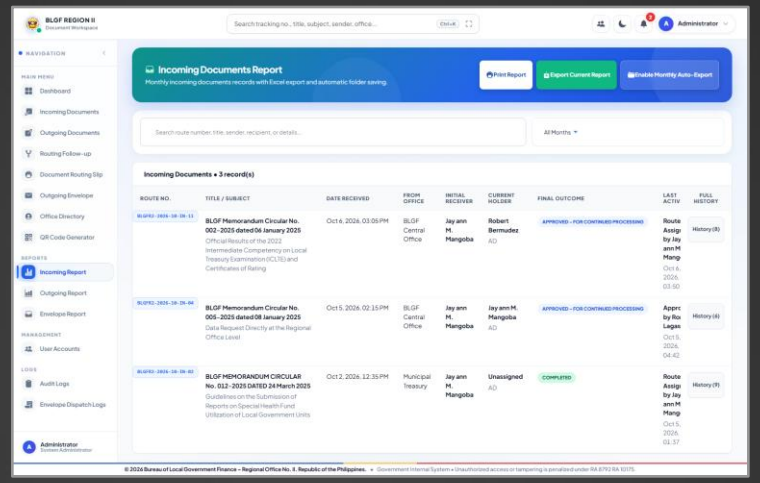
One permanent tracking number links document metadata, recipient routes, attachments, actions, status changes, and the complete transaction history.

SYSTEM DEMONSTRATION II

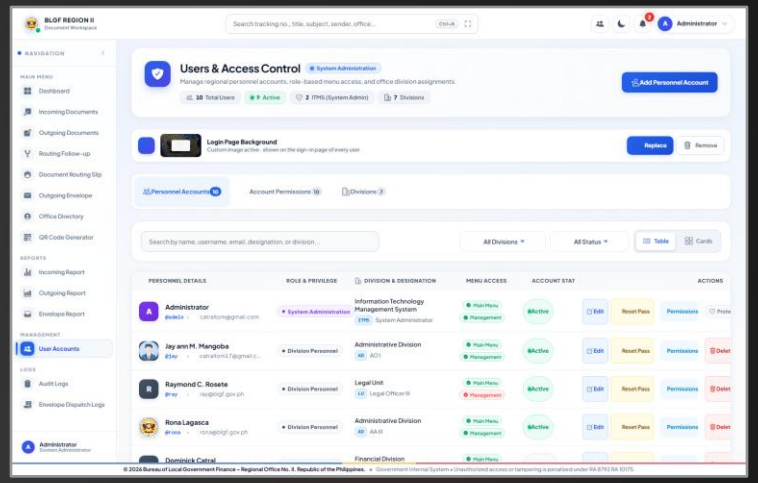
Monitoring, reporting, and administration



Routing Follow-up



Transaction Reports



User Accounts

Visibility and actions are controlled by role, account permissions, assignment, and route participation.

ISO/IEC 25010 EVALUATION

Actual survey administration and analysis

RESPONDENTS

23 total
13 BLGF employees
10 IT experts

INDICATORS

27 actual indicators
8 quality characteristics
5-point Likert scale

INTERPRETATION

4.21-5.00
VERY GREAT EXTENT
Weighted mean used

The reported results correspond to the digital questionnaire actually administered and reproduced in Appendix D.2.

ISO 25010 CATEGORY RESULTS

All characteristics were rated to a Very Great Extent



GRAND MEAN: 4.97 - VERY GREAT EXTENT

INDIVIDUAL INDICATOR HIGHLIGHTS

Chapter IV reports each indicator, not only the summary

PERFECT MEANS - 5.00

Required document-management functions
Workflow-task support
Data and transaction-history preservation
Documentation sufficiency
Straightforward task completion
Expected workload capacity
Role-based restriction
All four maintainability indicators

IMPROVEMENT PRIORITIES

Protection from unauthorized access/modification - 4.87
Efficient computing/network resource use - 4.91
Module information exchange - 4.91
Installation/deployment procedures - 4.91

All remained within Very Great Extent.

SUMMARY OF FINDINGS

- The existing manual workflow was vulnerable to delays, incomplete visibility, misplaced records, and weak traceability.
- The developed WDTMP provides end-to-end registration, routing, monitoring, retrieval, reporting, and auditing.
- All 27 indicators and all eight ISO/IEC 25010 characteristics were rated to a Very Great Extent.
- The overall evaluation produced a grand mean of 4.97; maintainability ranked highest at 5.00.
- Security remained highly rated at 4.93 but should receive continued review and strengthening.

CONCLUSIONS

The developed WDTMP addressed the identified workflow problems by consolidating document transactions into a secure, searchable, role-controlled, and auditable platform.

The 4.97 grand mean confirms that the system satisfies the evaluated ISO/IEC 25010 quality requirements to a Very Great Extent.

The platform is suitable for controlled operational adoption at BLGF Region II, subject to institutional approval, deployment preparation, and user training.

RECOMMENDATIONS

- Conduct controlled deployment with user orientation, role assignment, and documented operating procedures.
- Continue security testing, access reviews, backup verification, and audit-log monitoring.
- Maintain updated deployment documentation and monitor computing/network performance.
- Perform periodic user evaluation and incorporate approved workflow improvements.
- Consider future integrations only after privacy, interoperability, and institutional requirements are validated.

EXECUTIVE SUMMARY

PROBLEM

Manual routing caused delays, limited visibility, record risks, and weak accountability.

SOLUTION

WDTMP centralizes registration, routing, tracking, search, reporting, permissions, and audit trails.

EVALUATION

23 respondents rated 27 indicators. Grand mean: 4.97, interpreted as Very Great Extent.

DIRECTION

Proceed with controlled adoption while strengthening security, deployment guidance, and monitoring.

THANK YOU!

QUESTIONS AND DISCUSSION

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